

$N \log N$ Optimal Filtering

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1 Framework

The basic approach is outlined in Basu-Thakur Thesis Appendix E¹. Suppose a timestream $S(t)$ can be written as

$$S(t) = aA(t) + n(t), \quad (1)$$

where a is an amplitude factor, $A(t)$ is a signal, and $n(t)$ is noise. The spectral noise density of $n(t)$ can be written as $\tilde{J}(\nu) = \langle \tilde{n}^2(\nu) \rangle$. The estimator is given as

$$\chi^2(a) \equiv \sum_{\nu} \frac{|\tilde{S}(\nu) - a\tilde{A}(\nu)|^2}{\tilde{J}(\nu)}, \quad (2)$$

where $\tilde{S}(\nu)$ is the Fourier transform of $S(t)$ and $\tilde{A}(\nu)$ is the Fourier transform of $A(t)$. The best fit amplitude $\hat{a}$ is then given by

$$\hat{a} = \frac{\sum_{\nu} \tilde{S}(\nu) \tilde{A}^*(\nu) / \tilde{J}(\nu)}{\sum_{\nu} \tilde{A}(\nu) \tilde{A}^*(\nu) / \tilde{J}(\nu)}. \quad (3)$$

2 Discrete Optimal Filter for Signal Extraction

Now, suppose S_t is a discrete timestream of length L and $A_{t+\tau}$ is a timestream of length M , where τ is an offset that indicates the time where the signal occurs in S_t . $\hat{a}_{\tau}$ can then be interpreted as the amplitude of the signal found in S_t at time $t = \tau$. Over all values of τ in t , the $\hat{a}_{\tau}$ is expected to peak at times where the signal is found in S_t .

One can compute the optimal amplitudes $\hat{a}_{\tau}$ for all τ with

$$\hat{a}_{\tau} = \frac{\sum_{\nu} \tilde{S}_{\nu} \tilde{A}_{\nu}^* e^{-j2\pi\tau/L} / \tilde{J}_{\nu}}{\sum_{\nu} \tilde{A}_{\nu} \tilde{A}_{\nu}^* / \tilde{J}_{\nu}}. \quad (4)$$

The optimal filter is defined as

$$\tilde{H}_{\nu\tau} = \frac{\tilde{A}_{\nu}^* / \tilde{J}_{\nu}}{\sum_{\nu} \tilde{A}_{\nu} \tilde{A}_{\nu}^* / \tilde{J}_{\nu}} \quad (5)$$

such that

$$\hat{a}_{\tau} = \sum_{\nu} \tilde{S}_{\nu} \tilde{H}_{\nu\tau} e^{-j2\pi\tau/L}. \quad (6)$$

This is equivalent to the convolution of timestreams S_t and $H_{t\tau}$

$$\hat{a}_{\tau} = S_t * H_{t\tau}. \quad (7)$$

¹Ritoban Basu Thakur (Jan. 2014). “The Cryogenic Dark Matter Search low ionization-threshold experiment”. en. PhD thesis. University of Illinois at Urbana-Champaign, FERMILAB-THESIS-2014-33, 1182551. DOI: [10.2172/1182551](https://doi.org/10.2172/1182551). URL: <http://www.osti.gov/servlets/purl/1182551/> (visited on 06/16/2024).

3 Algorithm

Suppose the timestream S_t contains N instances of the signal A_t such that

$$S_t = \sum_k a_k A_{k,t+\tau_k} + n_t, \quad (8)$$

where $A_{k,t+\tau_k}$ represents the signal A_t offset by some time τ_k . Assume the values a_k are the expected to be the same for each instance of the signal up to a Gaussian noise term. The template A_t can be constructed from the sum of the signals with the time offset τ_k removed:

$$A_t = \sum_k a_k A_{k,t}. \quad (9)$$

The user can choose to normalize A_t such that the determined amplitudes a_k correspond to some physical value (for example, normalizing $\max A_t = 1$ ensures that a_k extracts the peak of each signal instance that is found).

The user starts with timestream S_t of length L . The optimal signal template A_t of length M is found iteratively using the following algorithm.

1. Construct an initial guess for A_t using Equation 9. The individual signals A_{kt} are found using conventional methods (e.g. peak finding in S_t).
2. Select the noise timestream J_t from S_t to be a region that does not contain any signal. Because the algorithm will use $\tilde{J}_\nu$, the user can select many such timestreams of length M and average to get $\tilde{J}_\nu$ for better results.
3. Calculate $\tilde{A}_\nu$ from the template A_t .
4. Calculate the optimal filter from Equation 5.
5. Calculate the inverse Fourier transform of the optimal filter to get $H_{t\tau}$.
6. Calculate $\hat{a}_\tau$ from Equation 7.
7. The peaks in $\hat{a}_\tau$ correspond to the optimal start times τ_k of the signal. Construct a new template using Equation 9. Return to step 3 and iterate until the template converges.

3.1 Time Complexity

Steps 1 and 2 are performed only a single time, and their complexity is dominated by the FFTs of the noise timestreams J_t , which scale as $\mathcal{O}(PM \log M)$, where P is the number of timestreams J_t that are selected. Steps 3-7 are iterated over a small number of times. Step 3 scales as $\mathcal{O}(M)$ and Steps 4 and 5 scale as $\mathcal{O}(M \log M)$. Step 6 can be performed with a FFT-based convolution, so it scales as $\mathcal{O}(L \log L)$. Step 7 scales as $\mathcal{O}(L)$. Therefore, the full algorithm time complexity is dominated by the $\mathcal{O}(L \log L)$ complexity of the convolution in step 6.